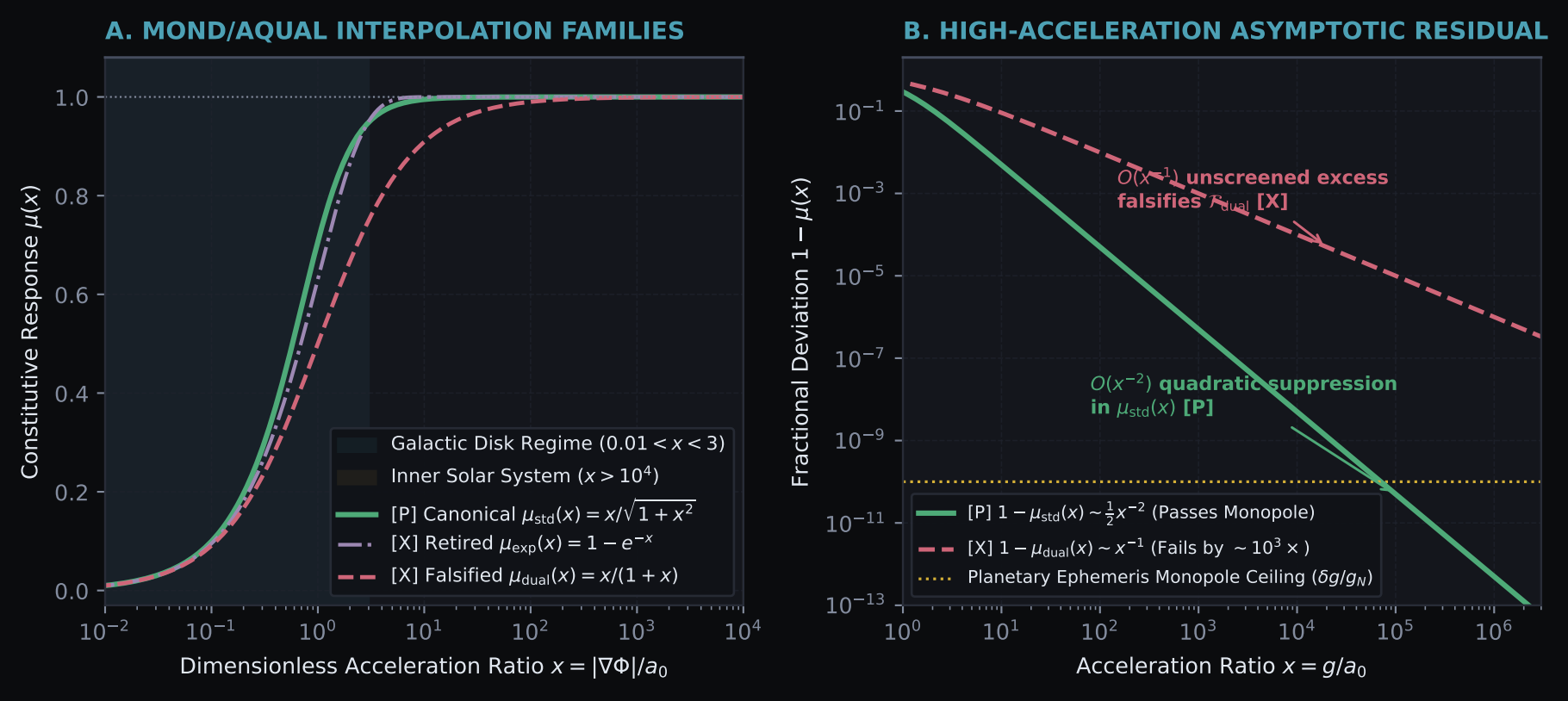
A. MOND/AQUAL INTERPOLATION FAMILIES
Constitutive Response $\mu(x)$
1.0
0.8
0.6
0.4
0.2
0.0
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
Dimensionless Acceleration Ratio $x = |\nabla\Phi|/a_0$
Galactic Disk Regime ($0.01 < x < 3$)
Inner Solar System ($x > 10^4$)
[P] Canonical $\mu_{\rm std}(x) = x/\sqrt{1+x^2}$
[X] Retired $\mu_{\rm exp}(x) = 1 - e^{-x}$
[X] Falsified $\mu_{\rm dual}(x) = x/(1+x)$
B. HIGH-ACCELERATION ASYMPTOTIC RESIDUAL
Fractional Deviation $1 - \mu(x)$
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}
$O(x^{-1})$ unscreened excess
falsifies $\mu_{\rm dual}$ [X]
$O(x^{-2})$ quadratic suppression
in $\mu_{\rm std}(x)$ [P]
[P] $1 - \mu_{\rm std}(x) \sim \frac{1}{2}x^{-2}$ (Passes Monopole)
[X] $1 - \mu_{\rm dual}(x) \sim x^{-1}$ (Fails by $\sim 10^3 \times$)
Planetary Ephemeris Monopole Ceiling ($\delta g/g_N$)
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
10^{5}
10^{6}
Acceleration Ratio $x = g/a_0$